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- C:\Users\umorrja\projects\testpapers\grade11_mathematical_literacy_papers\2020\MATHS LIT P1 GR11 MEMO NOV2020 ENG D.pdf
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- C:\Users\umorrja\projects\testpapers\grade11_mathematical_literacy_papers\2020\MATHS LIT P2 QP ADDEN MEMO GR11 NOV 2020 ENG D.pdf

--- Summary ---
of units

(4) F

L2

$$\begin{aligned} 2.1.3 \text{ \% loss} &= 12\,924,75 - 6\,995,25 \\ &= R5\,929,50 \\ &= 10075,1292450,5929u \\ &= 45,88\% \end{aligned}$$

OR

$$\begin{aligned} \text{Percentage loss} &= 10012924,7525,6995u \\ &= 54,12\% \\ &= 100\% - 54,12\% \\ &= 45,88\% \quad 1M \text{ Subtraction of values} \end{aligned}$$

1S Simplification

1M Dividing correct values

1M Multiply by 100%

OR

1M Dividing correct values 1M M

ultiply by 100%

1S Simplification

1M Subtraction of %

(4) F

L3

$$\begin{aligned} 2.1.4 B &= R8\,038,07 - R6\,995,25 \\ &= R1\,042,82 \quad 1MA \text{ Subtraction} \quad 1CA \text{ Correct answer} \end{aligned} \quad (2) F$$

L2

$$2.1.5 R765,57 \quad 2RT \text{ Correct value}$$

(2) F

L2

$$\begin{aligned} 2.1.6 \text{ \% increase} &= 10075,33275,33202,366u \blacksquare \\ &= 9,998\% \\ &= 10\% \quad 1RT \text{ Correct values} \end{aligned}$$

1SF Substitution

1S Simplification

1R Nearest % (4) F

L2

2.2.1 Number of plates 99 RT

2RT Number of plates

(2) F

L2

2.2.2 Fixed expenses = R500 99 RT 2RT Fixed expenses

(2) F

L2

2.2.3 Income = R50 × Number of plates sold 9 M 9 A 1M Multiplication with R50

1A Correct formula (2) F

L2

2.2.4 R0 OR (No Profit) 99 RT 2RT No profit

(2) F

L2 9 R 9 S 99 RT

9 M

9 S

9M 9 CA

9 M 9 C

9 CA

99RT 9C A 9 MA

9RT 9 M

9 M

9 SF 9 S 9 M

9 M

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2.2.5 Loss for 8 plates = Expenses – Income

9RT 9 RT

= 740 – 400 9M

= R340 9A

OR

Expenses = 500 + 8 × 30 = R740 9M

Income = 50 × 8 = R400 9M

Loss = 740 – 400 = R340 9A 1RT R740

1RT R400

1M Subtraction

1A Loss

(From graph allow 3 40±10)

OR

1M for R 740

1M for R400

1M subtraction

1A for R340 exact answer (4) F

L3

[30]

6 MATHEMATICAL LITERACY P1 (EC/NOVEMBER 2020)

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QUESTION 3: DATA HANDLING (18 marks) AND PROBABILITY (3 marks)

Ques.

Solution Explanation T/L

3.1 Total number of spas

= 30 394 + 13 856 + 3 984 + 6 057 + 46 282 + 48 679

=149 252 9M 9A

1M Adding correct values

1A Total (2) D

L1

3.2 Mean = ■■■■ ■■■■

■ 9M

= 24 875,33

= 24 875 9 R CA from 3.1

1M Division

1R Whole number
(2) D
L2

3.3 European spas as a % = ■■■ ■■■■
■■■■ ■■■■×100 9M
= 31% 9CA 1M Fraction with correct
values and
multiplication by 100
1CA Percentage (2) D
L2

3.4 Range = 48 679 – 3 984
= 44 695 9S
Number of regions above range = 2 CA 9
1S Calculate range
1CA Number of regions (2) D
L3

3.5 30 394 : 48 679 = 1 : ■■■ ■■■■
■■■ ■■■■ 9M 9M
= 1 : 1,60 9 CA 1M Ratio
1M Fraction
1CA Unit ratio
NPR (3) D
L3

3.6 Revenue in sub-saharan Africa = 6,6 – 5,0 9M
= 1,6 9S
Total revenue for spas
= 22,9 + 6,6 + 1,6 + 2,8 + 33,3 + 26,5 9M
= \$93,7 billion 9CA 1M Subtraction
1S Simplification
1M Addition
1CA Total revenue
Penalise 1 mark if not in
billions (4) D
L3

3.7 1,6; 2,8; 6,6; 22,9; 26,5; 33,3 9M

Median revenue = ■,■ ■■■,■
■ 9M
= \$14,75 billion 9CA CA the value \$1,6 from 3.6
included in the data
1M Arranging in order of
descending or ascending
1M Concept of median
1CA Answer in billions (3) D
L3

3.8 P (Regions with more than 40 000 spas)

$$\frac{\text{■}}{\text{■}} \times 100 \quad 9M$$

= 33,33% 9CA 1RT Correct numerator and denominator

1M Multiplication by 100

1CA

Percentage

NPR

(3) P

L2

[21]

9RT

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QUESTION 4: FINANCE (12 marks), DATA HANDLING (11 marks) AND
PROBABILITY (3 marks)

Ques.

≈ 0,091 CA vanaf 2.2.1

1A Aantal

verstelskroewe

1MCA

1R 3 des plekke (3) P

V2

2.2.3 Om skok te vermy

OF

Vir veiligheidsredes

Aanvaar enige ander relevante redes 2R Rede (2) M&P

V4

2.2.4 Gloeilampe kan breek/beskadig 2R Verduideliking

(2) M&P

V4

2.2.5 Om Stam 2 met Stam 3 te verbind 99A

2A Verduideliking

(2) M&P

V4 9C

9M

9CA

9SF

9MCA

9M

9A

9O

9SF

9MCA 9M

9CA

9R

99A

9A 9MCA

9R

99R

99R

4 WISKUNDIGE GELETTERDHEID V2 (EC/NOVEMBER 2020)

Kopiereg voorbehou Blaai om asseblief

VRAAG 3 [33 PUNTE]

Vr Oplossing Verduideliking O&V

3.1.1 Suidwes

Noord 2A Suidwes

1A Noord (4) M&P

V2

3.1.2 Skaal verwys na die verwantskap (verhouding) tussen die afstand op █ kaart en die ooreenstemmende afstand op die grond (in werklikheid) 2A Verduideliking

(2) M&P

V1

3.1.3 Lyn gemeet = 1,8 cm (Aanvaar 1,7 cm tot 1,9 cm)

1,8 cm = 2 km

1,8 cm = 200 000 cm

█ 1 cm = 111 111, 111

█ 1 : 111 000 1A Lyn gemeet

1C km na cm

1S Vereenvoudig 1R Naaste '000 (4) M&P

V3

3.1.4 Spoed = █

█

65 km/h = █, █ █

█

Tyd = █, █ █

█ █/█

= 0,28 ...h

= 16,8 min

= 17 minute

Tyd van aankoms = 14:53 + 17 minute

= 15:10

1SF Vervanging 1M

Verander

onderwerp van

formule

1A Tyd in uur

1CA T yd in min

1M Tel tye op

1CA Aankomstyd

(6) M&P

V3

3.1.5 As gevolg van die (Indiese) oseaan 2R Rede (2) M&P

V4

3.2.1 Biefstuk = 0,454 kg 2A Kilogram (2) M

V1

3.2.2 Sout = $0,5 \times 5$ ml

= 2,5 ml

Swart peper = $0,25 \times 5$ ml

= 1,25 ml

Totaal = 2,5 ml + 1,25 ml

= 3,75 ml

1MA Milliliter

1CA Milliliter

11CA Totaal (3) M

V2

3.2.3 1 koppie sous = 250 ml 9MA

█

█ koppie geklopte room = $0,75 \times 250$ ml

$$= 187,5 \text{ ml 9MA}$$

■

$$\begin{aligned} \blacksquare \text{ koppie suurroom} &= 0,25 \times 250 \\ &= 62,5 \text{ ml} \end{aligned}$$

$$\begin{aligned} \text{Totaal} &= 250 \text{ ml} + 187,5 \text{ ml} + 62,5 \text{ ml} \\ &= 500 \text{ ml} = 0,5 \text{ liter} \end{aligned}$$

Bewering is geldig 9O 1MA 250 ml
1MA 187,5 ml

1MA 62,5 ml

1CA Liter
1O Geldig

(5) M

V4
99A
9A
9C
9S
9R
9M
9A
9C
9M
9CA
99R 9M
99A
9MA
9CA
9CA

9MA
9CA 99A
99A

(EC/NOVEMBER 2020) WISKUNDIGE GELETTERDHEID V2 5
Kopiereg voorbehou Blaai om asseblief

3.2.4 Kooktyd = (3 min
× 2) + 8 min + 1 min + 1 min + 2 min + 20 min + 2
min
= 40 minute 1M Vermenigvuldig
met 2

1MCA Tel tye op 1CA Totale tyd (3) M
V2

3.2.5 Tyd vir voorbereiding moet ook in ag geneem word.

2A Rede (2) M

V4

VRAAG 4 [20 PUNTE]

Vr Oplossing Verduideliking O&V

4.1.1 6 sitplekke 2A Aantal sitplekke (2) M&P

V1

4.1.2 Verhoogde sitplekarea = 112 sitplekke Ge
lyke sitplekarea = 85 sitplekke
Verskil = 112 – 85

= 27 sitplekke 1A Sitplekke in
 Verhoogde area
 1A Sitplekke in
 Gelyke area
 1M Aftrekking
 1CA Verskil (4) M&P
 V3

4.1.3 P(L-ry) = ■■
 ■■■ 1A Teller
 1MCA Noemer
 (CA vanaf 4.1.2)
 (2) M&P
 V2

4.1.4 Loop tot by A5, draai dan regs
 Loop reguit verby Ry D
 Derde Ry (G) derde sitplek
 Aanvaar enige ander relevante antwoord 1A Reguit
 1A Draai regs
 1A 3de ry, 3de
 sitplek (3) M&P
 V4

4.2.1 Volume of drom = ■ × radius × radius × hoogte
 = 3,142 × 29 cm × 29 cm × 93 cm
 = 245 745, 246 cm³
 Liter = 245 745, 246 cm³ ÷ 1 000 cm³
 = 245,745 liter
 1C mm na cm
 1A Radius
 1SF Vervanging
 1CA Volume
 1CA Liter (5) M
 V3

4.2.2 x Die volume van die advertensie verwys na maksimum wat die
 drom kan hou.
 Solution Explanation Topic
 &
 Level
 3.1.1 Southwest
 North 2A Southwest
 1A North (4) M&P
 L2
 3.1.2 Scale refers to the relationship (ratio) between distance on a map
 and the corresponding distance on the ground (reality) 2A Definition (2) M&P
 L1

3.1.3 Measure bar = 1,8 cm (Accept 1,7 cm to 1,9 cm)
 1,8 cm = 2 km
 1,8 cm = 200 000 cm
 ■ 1 cm = 111 111, 111
 ■ 1 : 111 000 1A Measure bar
 1C km to cm

1S Simplification 1R Nearest '000 (4) M&P
 L3

3.1.4 Speed = ■■■■■■■■■■
 ■■■■
 65 km/h = ■■,■ ■■

■■■■■

Time = ■■■,■ ■■■

■■ ■■■/■

= 0,28h

= 16,8 min

= 17 minutes

Time of arrival = 14:53 + 17 minutes

= 15:10

1SF Substitution 1M Changing subject of formule 1A Time in hours 1CA Time in min 1M Add times

1CA Arrival time

(6) M&P

L3

3.1.5 Because of the (Indian) ocean 2R Reason

(2) M&P

L4

3.2.1

Steak = 0,454 kg

2A Kilogram (2) M

L1

3.2.2 Salt = $0,5 \times 5$ ml

= 2,5 ml

Black pepper = $0,25 \times 5$ ml

= 1,25 ml

Total = 2,5 ml + 1,25 ml

= 3,75 ml

1MA Millilitres 1CA Millilitres 11CA Total (3) M

L2

3.2.3 1 cup broth = 250 ml

■

■ cup whipping cream = $0,75 \times 250$ ml

= 187,5 ml

■

■ cup sour cream = $0,25 \times 250$

= 62,5 ml

Total = 250 ml + 187,5 ml + 62,5 ml

= 500 ml = 0,5 litres

Statement is valid 9O 1MA 250 ml

1MA 187,5 ml

1MA 62,5

ml

1CA Litre

1O Valid (5) M

L4 99A

99A

9A

9C

9S

9R

9M

9A

9C

9M

9CA

99R 9M

99A

9MA
 9CA
 9CA
 9MA
 9MA
 9CA 9MA 99A
 (EC/NOVEMBER 2020) MATHEMATICAL LITERACY P2 5
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3.2.4 Cooking time = (3 min
 $\times 2$) + 8 min + 1 min + 1 min + 2 min + 20 min + 2
 min
 = 40 minutes 1M Multiply by 2 1MCA Adding all
 times

1CA Total time (3) M
 L2

3.2.5 Time for preparation should also be considered.

Solution Explanation Topic

&

Level

3.1.1 Southwest

North 2A Southw est

1A North (4) M&P

L2

3.1.2 Scale refers to the relationship (ratio) between distance on a map
 and the corresponding distance on the ground (reality) 2A Definition (2) M&P
 L1

3.1.3 Measure bar = 1,8 cm (Accept 1,7 cm to 1,9 cm)

1,8 cm = 2 km

1,8 cm = 200 000 cm

■ 1 cm = 111 111, 111

■ 1 : 111 000 1A Measure bar

1C km to cm

1S Sim plification

1R Nearest '000 (4) M&P

L3

3.1.4 Speed = ■■■■■■■■

■■■■■

65 km/h = ■■,■ ■■

■■■■■

Time = 18,2 ■■

65 ■■/■

= 0,28....h

= 16,8 min

= 17 minutes

Time of arrival = 14:53 + 17 minutes

= 15:10

1SF Substitution

1M Changing

subject of formule

1A Time in hours

1CA Time in min

1M Add times

1CA Arrival time

(6) M&P

L3

3.1.5 Because of the (Indian) ocean 2R Reason (2) M&P
L4

3.2.1
Steak = 0,454 kg
2A Kilogram (2) M
L1

3.2.2 Salt = $0,5 \times 5$ ml
= 2,5 ml
Black pepper = $0,25 \times 5$ ml
= 1,25 ml
Total = 2,5 ml + 1,25 ml
= 3,75 ml
1MA Millilitres

1CA Millilitres

11CA Total (3) M
L2

3.2.3 1 cup broth = 250 ml
3
4 cup whipping cream = $0,75 \times 250$ ml
= 187,5 ml
1
4 cup sour cream = $0,25 \times 250$
= 62,5 ml
Total = 250 ml + 187,5 ml + 62,5 ml
= 500 ml = 0,5 litres
Statement is valid ☐ O 1MA 250 ml

1MA 187,5 ml

1MA 62,5 ml

1CA Litre
1O Valid (5) M
L4 ☐ ☐ ☐ A

☐ ☐ A

☐ A

☐ C

☐ S

☐ R

☐ M

☐ A

☐ C

☐ M

☐ CA

☐ ☐ R ☐ M

☐ ☐ A

☐ MA

☐ CA

☐ CA

☐ MA

☐ MA

☐ CA ☐ MA ☐ ☐ A

3.2.4 Cooking time
 = (3 min × 2) + 8 min + 1 min + 1 min + 2 min + 20 min + 2 min
 = 40 minutes 1M Multiply by 2
 1MCA Adding all times
 1CA Total time (3) M
 L2

3.2.5
 Time for preparation should also be considered.

--- Full Combined Text ---

NATIONAL
 SENIOR CERTIFICATE

GRADE 11

NOVEMBER 2020

MATHEMATICAL LITERACY P1
 MARKING GUIDELINE
 EXEMPLAR

MARKS: 100

Symbol Explanation

M Method

MA Method with accuracy

CA Consistent accuracy

A Accuracy

C Conversion

S Simplification

RT/RG/RM Reading from a table/Reading from a graph/Read from map

F Choosing the correct formula

SF Substitution in a formula

J Justification

P Penalty, e.g. for no units, incorrect rounding off etc.

R Rounding Off/Reason

AO Answer only

NPR No penalty for rounding

This marking guideline consists of 8 pages.

2 MATHEMATICAL LITERACY P1 (EC/NOVEMBER 2020)
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 MARKING GUIDELINES

NOTE:

- x If a candidate answers a question TWICE, only mark the FIRST attempt.
- x If a candidate has crossed out (cancelled) an attempt to a question and NOT redone the solution, mark the crossed out (cancelled version)
- x Consistent accuracy (CA) applies in ALL aspects of the marking guidelines, however it stops at the second calculation error.
- x If the candidate presents any extra solution when reading from a graph, table, layout plan and map, then penalise for every extra incorrect item presented.

LET WEL:

- x As ■ kandidaat ■ vraag TWEE keer beantwoord, merk slegs die EERSTE poging.
- x As ■ kandidaat ■ antwoord van ■ vraag doodtrek (kanselleer) en nie oordoen nie, merk die doodgetrekte (gekanselleerde) poging.
- x Volgehoue akkuraatheid (CA) word in ALLE aspekte van die nasienriglyn toegepas, maar dit hou by die tweede berekeningsfout op.
- x Wanneer ■ kandidaat aflesings vanaf ■ grafiek, tabel, uitlegplan en kaart geneem en ekstra antwoorde gee, penaliseer vir elke ekstra verkeerde item.

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QUESTION 1 [23 marks]

Ques. Solution Explanation T&L

$$1.1.1 \text{ Annual gross salary} = R10\,500 \times 12 \\ = R126\,000$$

1M Multiply by 12

1A Gross per annum (2) F

L1

$$1.1.2 \text{ Monthly food expense} = R10\,500 \times 36\% \\ = R3\,780 \text{ 1M \% Calculation}$$

1CA Amount (2) F

L1

1.1.3 Housing % : Food %

$$= 21\% : 36\%$$

$$= 7 : 12 \text{ 9CA}$$

1M Correct values and order 1C

A Simplest form (2) F

L1

$$1.1.4 \text{ Savings \%} = 100\% - (21\% + 36\% + 10\% + 1,9\%) \\ = 100\% - 68,9\% \\ = 31,1\% \text{ 1M Adding correct values}$$

1M Subtracting from 100

1CA Percentage (3) F

L1

1.2.1 Primary data

2A Correct data type

(2) D

L1

1.2.2 41 99 RT

2RT Highest mark

(2) D

L1

1.2.3 Median is the middle value of a set of data which is arranged from small to big. 2A Explanation

(2) D

L1

1.2.4 35

2A Correct mark

(2) D

L1

1.2.5 3 2RT No. of learners failed

(2) D

L1

1.3.1 Loss is when the cost is more than the income. 99 A
OR
Loss incurred when selling price is less than cost price of an item. 99 A 2A Correct explanation
(2) F

L1

1.3.2 % loss = ■■
■■■ × 100% 9M
= 6,67% 9 CA 1M Fraction multiplied by
100%

1CA Percentage

NPR (2) F

L1

[23]

9 M

9 A

9M

9M

99A

99 RT 9M

9CA

99 A 9M

9CA

99A

4 MATHEMATICAL LITERACY P1 (EC/NOVEMBER 2020)

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QUESTION 2: FINANCE [30 marks]

Ques. Solution Explanation Topic

/Level

2.1.1 SmartMAX Focussed Education Plan 1 2A Correct investment plan

(2) F

L1

2.1.2 Number of units =

100470, 8266

= R82,6647

= 6647,8207, 8038

= 97,23703104 1C Converted to Rands

1CA Value

1M Division

1CA No. of units

(4) F

L2

2.1.3 % loss = 12 924,75 – 6 995,25

= R5 929,50

= 10075, 1292450, 5929u

= 45,88%

OR

Percentage loss = 10012924,7525, 6995u

= 54,12%

= 100% – 54,12%

= 45,88% 1M Subtraction of values

1S Simplification

1M Dividing correct values

1M Multiply by 100%

OR

1M Dividing correct values 1M M
multiply by 100%
1S Simplification
1M Subtraction of %

(4) F

L3

2.1.4 B = R8 038,07 – R6 995,25
= R1 042,82 1MA Subtraction 1CA Correct answer (2) F

L2

2.1.5 R765,57 2RT Correct value
(2) F

L2

2.1.6 % increase = 10075,33275,33202,366u■
= 9,998%
= 10% 1RT Correct values

1SF Substitution

1S Simplification

1R Nearest % (4) F

L2

2.2.1 Number of plates 99 RT
2RT Number of plates

(2) F

L2

2.2.2 Fixed expenses = R500 99 RT 2RT Fixed expenses
(2) F

L2

2.2.3 Income = R50 × Number of plates sold 9 M 9 A 1M Multiplication with R50
1A Correct formula (2) F

L2

2.2.4 R0 OR (No Profit) 99 RT 2RT No profit
(2) F

L2 9 R 9 S 99 RT

9 M

9 S

9M 9 CA

9 M 9 C

9 CA

99RT 9C A 9 MA

9RT 9 M

9 M

9 SF 9 S 9 M

9 M

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2.2.5 Loss for 8 plates = Expenses – Income
9RT 9 RT
= 740 – 400 9M
= R340 9A

OR

Expenses = 500 + 8 × 30 = R740 9M

Income = 50 × 8 = R400 9M

Loss = 740 – 400 = R340 9A 1RT R740

1RT R400

1M Subtraction

1A Loss
 (From graph allow 340 ± 10)
 OR
 1M for R 740
 1M for R400
 1M subtraction
 1A for R340 exact answer (4) F
 L3

[30]

6 MATHEMATICAL LITERACY P1 (EC/NOVEMBER 2020)
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 QUESTION 3: DATA HANDLING (18 marks) AND PROBABILITY (3 marks)
 Ques. Solution Explanation T/L

3.1 Total number of spas
 $= 30\,394 + 13\,856 + 3\,984 + 6\,057 + 46\,282 + 48\,679$
 $= 149\,252$ 9M 9A
 1M Adding correct values
 1A Total (2) D
 L1

3.2 Mean = $\frac{149\,252}{6}$
 $= 24\,875,33$
 $= 24\,875,9$ R CA from 3.1
 1M Division
 1R Whole number
 (2) D
 L2

3.3 European spas as a % = $\frac{149\,252}{486\,790} \times 100$ 9M
 $= 31\%$ 9CA 1M Fraction with correct
 values and
 multiplication by 100
 1CA Percentage (2) D
 L2

3.4 Range = $48\,679 - 3\,984$
 $= 44\,695$ 9S
 Number of regions above range = 2 CA 9
 1S Calculate range
 1CA Number of regions (2) D
 L3

3.5 $30\,394 : 48\,679 = 1 : 1,609$ CA 1M Ratio
 $= 1 : 1,609$ CA 1M Ratio
 1M Fraction
 1CA Unit ratio
 NPR (3) D
 L3

3.6 Revenue in sub-saharan Africa = $6,6 - 5,0$ 9M
 $= 1,6$ 9S

Total revenue for spas
 $= 22,9 + 6,6 + 1,6 + 2,8 + 33,3 + 26,5$ 9M
 $= \$93,7$ billion 9CA 1M Subtraction

1S Simplification

1M Addition

1CA Total revenue

Penalise 1 mark if not in
billions (4) D

L3

3.7 1,6; 2,8; 6,6; 22,9; 26,5; 33,3 9M

Median revenue = ■, ■ ■ ■ ■, ■

■ 9M

$= \$14,75$ billion 9CA CA the value \$1,6 from 3.6

included in the data

1M Arranging in order of
descending or ascending

1M Concept of median

1CA Answer in billions (3) D

L3

3.8 P (Regions with more than 40 000 spas)

$= \frac{\text{■}}{\text{■}}$

■ $\times 100$ 9M

$= 33,33\%$ 9CA 1RT Correct numerator and
denominator

1M Multiplication by 100

1CA

Percentage

NPR (3) P

L2

[21]

9RT

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QUESTION 4: FINANCE (12 marks), DATA HANDLING (11 marks) AND
PROBABILITY (3 marks)

Ques. Solution Explanation T/L

4.1 Cost =

Kilolitre Cost

$6 \times 0 = 0$ 9M

$9 \times 9,35 = R84,15$

$10 \times 11,16 = R111,6$

Total = 25 litres $84,15 + 111,60 = R195,75$

$R195 \times 115\% = R225,11$

OR

Cost = $(6 \times 0) + (9 \times 9,35) + (10 \times 11,16)$

$= R84,15 + R111,60$

$= R195,75$

Including VAT = $R195,75 \times 15\%$

$= R29,3625$

$= R195,75 + R29,3625$

$$= R225,11$$

Increased block rate tariffs to encourage saving of water 9A
OR

Also assist small businesses or families with free water 9A

Accept any other sound reason .

1M

Cost of first 6 k ■

1M Cost for both 9 and 10

kilolitres

1CA Total cost

1M Multiply by 15%

1CA Cost including VAT

1A

Reason

(6) F

L4

4.2 R0,019 = 1 RWF

R? = 745 614,04 RWF 9M

R? = 0,019 × 745 614,04 9M

= R14 166,66676 9S

Bank charges = 14166,66676 × ■■

■■■ 9M

= R1 416,66676 9A

Andile received = R14 166,66676 – 1 416,66676 9 M

= R12 750

Statement is valid. 9A

OR

Bank charges = ■■

■■■ × 745 614,04 RWF 9M

= 74 561,404 9A

Andile received in RWF = 745 614,04 – 74 561,404 9M

= 671 052,636 9S

In Rands: R0,019 = 1 RWF

R? = 671 052,636 RWF 9M

Andile received = R0,019 × 671 052,636 9M

= R12 750

Statement is valid 9CA 1M Concept of ratio

1M

Multiplication

1S Simplification value in R

1M Multiplication of 10%

1A Value of 10%

1M Subtraction

1A

Valid

OR

1M Multiplication of 10%

1A Value of 10%

1M Subtraction
1S Simplification value
1M
Concept of ratio
1M Multiplication

1CA Valid (6) F
L4

9CA 9M
9M 9M 9M 9M
9M
9CA 9CA
8 MATHEMATICAL LITERACY P1 (EC/NOVEMBER 2020)
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4.3.1 Total absentees = 67 9 M
Absentees on Wednesday = 16 9A
 $P(\text{absent on Wed}) = \frac{16}{67}$
 $\frac{16}{67}$ 9CA 1M Addition to 67
1A Absentees on Wed
1CA Fraction
(3) P
L2

4.3.2

1A for Monday

boys at 5
1A for Tuesday girls at 6
1A for Thurs for girls at 7
1A for Thurs for boys at 9
1A for Fri for boys at 7

(5) D
L2

4.4.1 Value of C:
 $64,2 = \frac{1}{30} C + 1,853$
 $\frac{1}{30} C$ 9M
 $64,2 \times 30 = C + 1,853$
 $C = 1,926 - 1,853$ 9M
 $= 0,073$ 9CA
Answer invalid
1M Addition (1,853) and
division by 30

1M Subtraction
1CA Value of C

1A Invalid (4) D
L4

4.4.2 D = 0 9A

No learners scored 30 – 39 marks 9A 1A Value of D

1A Explanation

(CA value of D from 4. 4.1

included in the data) (2) D

L4

[26]

TOTAL: 100

024681012

Monday Tuesday Wednesday Thursday Friday Number of absent learners

Days of the week Number of absent learners during the week

Girls Boys 9A

9A

Tuesday

D9A

9O 9A 9A

NASIONALE

SENIOR SERTIFIKAAT

GRAAD 11

NOVEMBER 2020

WISKUNDIGE GELETTERDHEID V2

NASIENRIGLYN

(EKSEMPLAAR)

PUNTE: 100

Hierdie nasien riglyn bestaan uit 5 bladsye.

Simbool Verduideliking

M Metode

MA Metode met akkuraatheid

MCA Metode met deurlopende akkuraatheid

CA Deurlopende akkuraatheid

A Akkuraatheid

C Omskakeling

S Vereenvoudiging

RT/RG/RM Lees vanaf ■ tabel/Lees vanaf ■ grafiek Lees vanaf ■ kaart

F Kies die korrekte formule

SF Vervanging in ■ formule

O Opinie/Verduideliking

P Penaliseer, bv. Vir geen eenhede, foutiewe ronding ens.

R Ronding/Rede

AO Slegs antwoord

NPR Geen penalisering vir ronding

2 WISKUNDIGE GELETTERDHEID V2 (EC/NOVEMBER 2020)

Kopiereg voorbehou Blaai om asseblief

VRAAG 1 [21 PUNTE]

Vr Oplossing Verduideliking O&V

1.1.1 Temperatuurlesing = 30°C 2A Korrekte lesing (2) M
V1

1.1.2 $^{\circ}\text{F} = 100^{\circ}\text{F}$ 2A $^{\circ}\text{F}$ (2) M
V1

1.1.3 Minimum $^{\circ}\text{Fahrenheit}$ = minus 42°F OF - 42°F
2A Min $^{\circ}\text{F}$ (2) M
V1

1.1.4 Termometer 1 : Termometer 2 =
 $90 : 20$
= $9 : 2$ 1RD Beide waardes korrek

1M Korrekte orde
1CA
Vereenvoudigde verhouding (3) M
V1

1.2.1 Aantal huise = 18 2A Korrekte aantal huise (2) M
V1

1.2.2 Hospitaal 2A Korrekte gebou (2) M
V1

1.2.3 Kafee 2A Korrekte besigheid (2) M
V1

1.2.4 Lengte van die balk = $1,5\text{ cm}$ 2A Korrekte lengte
(2) M
V1

1.2.5 $1,5\text{ cm}$ op die kaart verteenwoordig 50 jaart in werklikheid CA vanaf 1.2.4
2A Verduidelik ing (2) M
V1

1.2.6 3 huise 2A Aantal meer huise (2) M
V1 99A

99A

99A

99A

99A

99A

99A

99A 9RD

9M

9CA

99A

(EC/NOVEMBER 2020) WISKUNDIGE GELETTERDHEID V2 3

Kopiereg voorbehou Blaai om asseblief

VRAAG 2 [26 PUNTE]

Vr. Oplossing Verduideliking O&V

2.1.1 Omtrek is die afstand of lengte rondom die lamskerm 2A Verduideliking
(2) M
V1

2.1.2 Radius = ■■■■

■■■

= ■■■

■

= 10 cm 1C mm na cm 1M

Deel deur 2

1CA Radius (3) M

V2

2.1.3 Omtrek van lampskerm = ■ × d e u r s n e e

= 3,142 × 20 cm

= 62,84 cm CA vanaf 2.1.1

1SF Vervanging

1MCA Omtrek (2) M

V2

2.1.4 Hoogte van lampskerm = 20 cm × 1,65

= 33 cm

Bewering nie geldig nie 1M

Vermenigvuldig

met 1,65

1A Hoogte

1O

Nie geldig nie (3) M

V4

2.1.5 Oppervlakte van lampskerm = ■ × r a d i u s × r a d i u s

= 3,142 × 10 cm × 10 cm

= 314, 2 cm²

Oppervlakte insluitende vermorsing = 314, 2 cm² × 1,0425

= 327,5535 cm²

≈ 328 cm² CA vanaf 2.1.1

1SF Vervanging

1MCA Opp van lampskerm

1M

Vermenigvuldig met

1,0425

1CA Opp

insluitend

vermorsing

1R Naaste cm² (5) M

V3

2.2.1 Aantal dele = 22 dele 2A Aantal dele

(2) M&P

V2

2.2.2 P(verstelskroef) = ■

■■

= 0,090909....

≈ 0,091 CA vanaf 2.2.1

1A Aantal

verstelskroewe

1MCA

1R 3 des plekke (3) P

V2

2.2.3 Om skok te vermy

OF

Vir veiligheidsredes

Aanvaar enige ander relevante redes 2R Rede (2) M&P

V4

2.2.4 Gloeilampe kan breek/beskadig 2R Verduideliking

(2) M&P

V4

2.2.5 Om Stam 2 met Stam 3 te verbind 99A

2A Verduideliking

(2) M&P

V4 9C

9M

9CA

9SF

9MCA

9M

9A

9O

9SF

9MCA 9M

9CA

9R

99A

9A 9MCA

9R

99R

99R

4 WISKUNDIGE GELETTERDHEID V2 (EC/NOVEMBER 2020)

Kopiereg voorbehou Blaai om asseblief

VRAAG 3 [33 PUNTE]

Vr Oplossing Verduideliking O&V

3.1.1 Suidwes

Noord 2A Suidwes

1A Noord (4) M&P

V2

3.1.2 Skaal verwys na die verwantskap (verhouding) tussen die afstand op ■ kaart en die ooreenstemmende afstand op die grond (in werklikheid) 2A Verduideliking

(2) M&P

V1

3.1.3 Lyn gemeet = 1,8 cm (Aanvaar 1,7 cm tot 1,9 cm)

1,8 cm = 2 km

1,8 cm = 200 000 cm

■ 1 cm = 111 111, 111

■ 1 : 111 000 1A Lyn gemeet

1C km na cm

1S Vereenvoudig 1R Naaste '000 (4) M&P

V3

3.1.4 Spoed = ■■■■■■■■

■■■

65 km/h = ■■,■ ■■

■■■

Tyd = ■■,■ ■■

■■ ■■/■

= 0,28h

= 16,8 min

= 17 minute

Tyd van aankoms = 14:53 + 17 minute

= 15:10

1SF Vervanging 1M

Verander

onderwerp van

formule

1A Tyd in uur

1CA T yd in min
1M Tel tye op
1CA Aankomstyd
(6) M&P
V3

3.1.5 As gevolg van die (Indiese) oseaan 2R Rede (2) M&P
V4

3.2.1 Biefstuk = 0,454 kg 2A Kilogram (2) M
V1

3.2.2 Sout = $0,5 \times 5$ ml
= 2,5 ml
Swart peper = $0,25 \times 5$ ml
= 1,25 ml
Totaal = 2,5 ml + 1,25 ml
= 3,75 ml
1MA Milliliter

1CA Milliliter

11CA Totaal (3) M
V2

3.2.3 1 koppie sous = 250 ml 9MA

■
■ koppie geklopte room = $0,75 \times 250$ ml
= 187,5 ml 9MA

■
■ koppie suurroom = $0,25 \times 250$
= 62,5 ml
Totaal = 250 ml + 187,5 ml + 62,5 ml
= 500 ml = 0,5 liter
Bewering is geldig 9O 1MA 250 ml
1MA 187,5 ml

1MA 62,5 ml

1CA Liter
1O Geldig

(5) M

V4
99A
9A
9C
9S
9R
9M
9A
9C
9M
9CA
99R 9M
99A
9MA
9CA
9CA

9MA
9CA 99A
99A
(EC/NOVEMBER 2020) WISKUNDIGE GELETTERDHEID V2 5
Kopiereg voorbehou Blaai om asseblief

3.2.4 Kooktyd = (3 min
× 2) + 8 min + 1 min + 1 min + 2 min + 20 min + 2
min
= 40 minute 1M Vermenigvuldig
met 2

1MCA Tel tye op 1CA Totale tyd (3) M
V2

3.2.5 Tyd vir voorbereiding moet ook in ag geneem word. 2A Rede (2) M
V4

VRAAG 4 [20 PUNTE]

Vr Oplossing Verduideliking O&V

4.1.1 6 sitplekke 2A Aantal sitplekke (2) M&P
V1

4.1.2 Verhoogde sitplekarea = 112 sitplekke Ge
lyke sitplekarea = 85 sitplekke
Verskil = 112 – 85
= 27 sitplekke 1A Sitplekke in

Verhoogde area

1A Sitplekke in

Gelyke area

1M Aftrekking

1CA Verskil (4) M&P

V3

4.1.3 $P(L-ry) = \blacksquare$

$\blacksquare$ 1A Teller

1MCA Noemer

(CA vanaf 4.1.2)

(2) M&P

V2

4.1.4 Loop tot by A5, draai dan regs

Loop reguit verby Ry D

Derde Ry (G) derde sitplek

Aanvaar enige ander relevante antwoord 1A Reguit

1A Draai regs

1A 3de ry, 3de

sitplek (3) M&P

V4

4.2.1 Volume of drom = $\blacksquare \times \text{radius} \times \text{radius} \times \text{hoogte}$

$$= 3,142 \times 29 \text{ cm} \times 29 \text{ cm} \times 93 \text{ cm}$$

$$= 245\,745,246 \text{ cm}^3$$

$$\text{Liter} = 245\,745,246 \text{ cm}^3 \div 1\,000 \text{ cm}^3$$

$$= 245,745 \text{ liter}$$

1C mm na cm

1A Radius

1SF Vervanging

1CA Volume

1CA Liter (5) M

V3

4.2.2 x Die volume van die advertensie verwys na maksimum wat die drom kan hou. 99A
 x Die berekende volume verwys na die drom in sy geheel. 99A 2A 1ste Rede 2A 2
 de Rede (4) M
 V4
 TOTAAL: 100 9A 9A
 9A
 9CA
 9A
 9A 9A
 9MCA
 9A
 9SF 9C
 9CA
 9CA
 9MCA
 9CA
 99A 9M

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GRADE 11

NOVEMBER 2020

MATHEMATICAL LITERACY P2
 MARKING GUIDELINE
 (EXEMPLAR)

MARKS: 100

This marking guideline consists of 5 pages .
 Symbol Explanation
 M Method
 M/A Method with Accuracy
 MCA Method with Consistent Accuracy
 CA Consistent Accuracy
 A Accuracy
 C Conversion
 S Simplification
 RT/RG/RM Reading from a table OR Reading from a graph OR Read from map
 F Choosing the correct formula
 SF Substitution in a formula
 J Justification
 P Penalty, e.g. for no units, incorrect rounding off etc.
 R Rounding Off OR Reason
 AO Answer only
 NPR No penalty for rounding
 2 MATHEMATICAL LITERACY P2 (EC/NOVEMBER 2020)
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QUESTION 1 [21 MARKS]
 Ques. Solution Explanation Topic
 &
 Level

1.1.1 Temperature reading = 30 °C 2A Correct reading
 M

(2)

L1

1.1.2 $^{\circ}\text{F} = 100^{\circ}\text{F}$ 2A $^{\circ}\text{F}$ (2) M

L1

1.1.3 Minimum $^{\circ}\text{Fahrenheit}$ = minus 42°F OR - 42°F
2A Min $^{\circ}\text{F}$ (2) M

L1

1.1.4 Thermometer 1 : Thermometer 2

=

90 : 20

= 9 : 2 1RD Both values correct

1M Correct order 1CA Simplified ratio (3) M

L1

1.2.1 Number of houses = 18 2A Correct number of houses (2) M

L1

1.2.2 Hospital 2A Correct building (2) M

L1

1.2.3 Café 2A Correct business (2) M

L1

1.2.4 Length of the bar = 1,5 cm 2A Correct length (2)

M

L1

1.2.5 1,5 cm on the map represents 50 yards in reality CA from 1.2.4

2A Explanation (2) M

L1

1.2.6 3 houses 2A Number of houses more (2) M

L1

[21] 99A

99A

99A

99A

99A

99A

99A

99A 9RD

9M

9CA

99A

(EC/NOVEMBER 2020) MATHEMATICAL LITERACY P2 3

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QUESTION 2 [26 MARKS]

Ques. Solution Explanation Topic

&

Level

2.1.1 Circumference is the distance or length around the lamp
shade 99A 2A Explanation (2) M

L1

2.1.2 Radius = ■■■■

■■■

= ■■■

■

= 10 cm 1C mm to cm

1M Divide by 2 1CA Radius (3) M

L2

$$\begin{aligned}
 2.1.3 \text{ Circumference of lamp shade} &= \pi \times \text{d i a m e t e r} \\
 &= 3,142 \times 20 \text{ cm} \\
 &= 62,84 \text{ cm} \quad \text{CA from 2.1.1}
 \end{aligned}$$

1SF Substitution

1MCA

Circumference (2) M

L2

$$\begin{aligned}
 2.1.4 \text{ Height of lamp shade} &= 20 \text{ cm} \times 1,65 \\
 &= 33 \text{ cm}
 \end{aligned}$$

Statement invalid 1M Multiply by

1,65

1A Height

1O Invalid (3) M

L4

$$\begin{aligned}
 2.1.5 \text{ Area of lampshade} &= \pi \times r \text{ a d i u s} \times r \text{ a d i u s} \\
 &= 3,142 \times 10 \text{ cm} \times 10 \text{ cm} \\
 &= 314,2 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Area including wastage} &= 314,2 \text{ cm}^2 \times 1,0425 \\
 &= 327,5535 \text{ cm}^2 \\
 &\approx 328 \text{ cm}^2 \quad \text{CA from 2.1.1}
 \end{aligned}$$

1SF Substitution

1MCA Area of lamp shade

1M Multiply by

1,0425

1CA Area including

wastage

1R Nearest cm² (5) M

L3

$$\begin{aligned}
 2.2.1 \text{ Number of parts} &= 22 \text{ parts} \quad 2A \text{ Number of parts} \\
 &\quad (2) \text{ M\&P}
 \end{aligned}$$

L2

$$2.2.2 \text{ P(set screw)} = \pi$$

π

$$= 0,090909....$$

$$\approx 0,091 \quad \text{CA from 2.2.1}$$

1A Number of set

screws

1MCA

1R 3 dec places (3) P

L2

2.2.3 To prevent shocking

OR

For safety purposes

Accept any other relevant reasons 2R Reason (2) M&P

L4

2.2.4 Bulbs can break/damage 2R Explanation (2) M&P

L4

2.2.5 To connect Stem 2 with Stem 3 2A Explanation (2) M&P

L4 9C

9M

9CA

9SF

9MCA

9M

9A

9O
9SF
9MCA
9M
9CA
9R
99A
9A
9MCA
9R
99R
99R
99A

4 MATHEMATICAL LITERACY P2 (EC/NOVEMBER 2020)
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QUESTION 3 [33 MARKS]
Ques. Solution Explanation Topic
&
Level
3.1.1 Southwest
North 2A Southwest
1A North (4) M&P
L2

3.1.2 Scale refers to the relationship (ratio) between distance on a map
and the corresponding distance on the ground (reality) 2A Definition (2) M&P
L1

3.1.3 Measure bar = 1,8 cm (Accept 1,7 cm to 1,9 cm)
1,8 cm = 2 km
1,8 cm = 200 000 cm
■ 1 cm = 111 111, 111
■ 1 : 111 000 1A Measure bar
1C km to cm

1S Simplification 1R Nearest '000 (4) M&P
L3

3.1.4 Speed = ■■■■■■■■
■■■■■

65 km/h = ■■,■ ■■
■■■■■

Time = ■■,■ ■■
■■ ■■/■

= 0,28h
= 16,8 min
= 17 minutes

Time of arrival = 14:53 + 17 minutes
= 15:10

1SF Substitution 1M Changing subject of formule 1A Time in hours 1CA Time in min 1M Add
times

1CA Arrival time
(6) M&P
L3

3.1.5 Because of the (Indian) ocean 2R Reason (2) M&P
L4

3.2.1
Steak = 0,454 kg
2A Kilogram (2) M

L1

$$3.2.2 \text{ Salt} = 0,5 \times 5 \text{ ml} \\ = 2,5 \text{ ml}$$

$$\text{Black pepper} = 0,25 \times 5 \text{ ml} \\ = 1,25 \text{ ml}$$

$$\text{Total} = 2,5 \text{ ml} + 1,25 \text{ ml} \\ = 3,75 \text{ ml}$$

1MA Millilitres 1CA Millilitres 11CA Total (3) M
L2

$$3.2.3 \text{ 1 cup broth} = 250 \text{ ml}$$

■

$$\text{■ cup whipping cream} = 0,75 \times 250 \text{ ml} \\ = 187,5 \text{ ml}$$

■

$$\text{■ cup sour cream} = 0,25 \times 250 \\ = 62,5 \text{ ml}$$

$$\text{Total} = 250 \text{ ml} + 187,5 \text{ ml} + 62,5 \text{ ml} \\ = 500 \text{ ml} = 0,5 \text{ litres}$$

Statement is valid 9O 1MA 250 ml
1MA 187,5 ml

1MA 62,5
ml

1CA Litre
1O Valid (5) M

L4 99A

99A

9A

9C

9S

9R

9M

9A

9C

9M

9CA

99R 9M

99A

9MA

9CA

9CA

9MA

9MA

9CA 9MA 99A

(EC/NOVEMBER 2020) MATHEMATICAL LITERACY P2 5

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$$3.2.4 \text{ Cooking time} = (3 \text{ min} \\ \times 2) + 8 \text{ min} + 1 \text{ min} + 1 \text{ min} + 2 \text{ min} + 20 \text{ min} + 2 \\ \text{min} \\ = 40 \text{ minutes} \quad 1M \text{ Multiply by 2 } 1MCA \text{ Adding all} \\ \text{times}$$

1CA Total time (3) M
L2

3.2.5 Time for preparation should also be considered. 2A Reason (2) M
L4

QUESTION 4 [20 MARKS]

Ques. Solution Explanation Topic

&

Level

4.1.1 6 seats 2A No of seats

(2) M&P

L1

4.1.2 Raised Seating Area = 112 seats Flat Seating Area = 85 seats

Difference = 112 – 85

= 27 seats 1A Seats in RSA

1A Seats in FSA 1M Subtract

1CA Difference (4) M&P

L3

4.1.3 P(L-row) = ■■■

■■■■ 1A Numerator

1MCA Denominator (CA

from 4.1.2) (2) M&P

L2

4.1.4 Walk to A5, then turn right

Walk straight pass the Row D

Third Row (G) third seat

Accept any other relevant responses 1A Straight

1A Turn right

1A 3rd row, 3rd seat

(3) M&P

L4

4.2.1 Volume of drum = ■ × radius × radius × height

= 3,142 × 29 cm × 29 cm × 93 cm

= 245 745, 246 cm³

Litres = 245 745, 246 cm³ ÷ 1 000 cm³

= 245,745 litres

1C mm to cm

1A Radius

1SF Substitution

1CA Volume

1CA Litres (5) M

L3

4.2.2 x The volume of advertisement refers to the maximum the drum can hold. 99A

x The calculated volume refers to the entire drum. 2A 1st Reason

2A 2

nd Reason (4) M

L4

TOTAL: 100 9A 99A

9A

9A

9M

9CA

9A

9A 9A

9MCA 9M 9MCA

9CA

99A

9A

9SF 9C

9CA

9CA

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GRADE 11

NOVEMBER 2020

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P Penalty, e.g. for no units, incorrect rounding off etc.

R Rounding Off OR Reason

AO Answer only

NPR No penalty for rounding

2 MATHEMATICAL LITERACY P2 (EC/NOVEMBER 2020)

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QUESTION 1 [21 MARKS]

Ques. Solution Explanation Topic

&

Level

1.1.1 Temperature reading = 30°C 2A Correct reading

(2) M

L1

1.1.2 $^{\circ}\text{F} = 100^{\circ}\text{F}$ 2A $^{\circ}\text{F}$ (2) M

L1

1.1.3 Minimum $^{\circ}\text{Fahrenheit}$ = minus 42°F OR -42°F

2A Min $^{\circ}\text{F}$ (2) M

L1

1.1.4 Thermometer 1 : Thermometer 2

= $90 : 20$

= $9 : 2$ 1RD Both values

correct

1M Correct order

1CA Simplified

ratio (3) M

L1

1.2.1 Number of houses = 18 2A Correct number
of houses (2) M
L1

1.2.2 Hospital
2A Correct building
(2) M
L1

1.2.3 Café 2A Correct business
(2) M
L1

1.2.4 Length of the bar = 1,5 cm 2A Correct length
(2) M
L1

1.2.5 1,5 cm on the map represents 50 yards in reality CA from 1.2.4
2A Explanation (2) M
L1

1.2.6 3 houses 2A Number of
houses more (2) M
L1

[21] ■■A

■■A

■■A

■■A

■■A

■■A

■■A

■■A ■RD

■M

■CA

■■A

(EC/NOVEMBER 2020) MATHEMATICAL LITERACY P2 3

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QUESTION 2 [26 MARKS]

Ques. Solution Explanation Topic
&

Level

2.1.1 Circumference is the distance or length around the lamp
shade ■■A 2A Explanation (2) M

L1

2.1.2 Radius = 200

10

= 20

2

= 10 cm 1C mm to cm

1M Divide by 2

1CA Radius (3) M

L2

2.1.3 Circumference of lamp shade = ■× diameter
= 3,142 × 20 cm
= 62,84 cm CA from 2.1.1

1SF Substitution

1MCA

Circumference (2) M

L2

$$2.1.4 \text{ Height of lamp shade} = 20 \text{ cm} \times 1,65 \\ = 33 \text{ cm}$$

Statement invalid 1M Multiply by 1,65

1A Height

1O Invalid (3) M

L4

$$2.1.5 \text{ Area of lampshade} = \pi \times \text{radius} \times \text{radius} \\ = 3,142 \times 10 \text{ cm} \times 10 \text{ cm} \\ = 314,2 \text{ cm}^2$$

$$\text{Area including wastage} = 314,2 \text{ cm}^2 \times 1,0425 \\ = 327,5535 \text{ cm}^2 \\ \approx 328 \text{ cm}^2 \text{ CA from 2.1.1}$$

1SF Substitution

1MCA Area of

lamp shade

1M Multiply by

1,0425

1CA Area including wastage

1R Nearest cm² (5) M

L3

$$2.2.1 \text{ Number of parts} = 22 \text{ parts} \quad 2A \text{ Number of parts} \\ (2) \text{ M\&P}$$

L2

$$2.2.2 \text{ P(set screw)} = 2 \\ 22$$

$$= 0,090909....$$

$$\approx 0,091 \text{ CA from 2.2.1}$$

1A Number of set

screws

1MCA

1R 3 dec places (3) P

L2

2.2.3 To prevent shocking

OR

For safety purposes

Accept any other relevant reasons 2R Reason (2) M&P

L4

2.2.4 Bulbs can break/damage 2R Explanation (2) M&P

L4

2.2.5 To connect Stem 2 with Stem 3 2A Explanation (2) M&P

L4 ■C

■M

■CA

■SF

■MCA

■M

■A

■O

■SF

■MCA

■M

■CA

■R

■■A

■A

■MCA

■R
■■R
■■R
■■A

4 MATHEMATICAL LITERACY P2 (EC/NOVEMBER 2020)

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QUESTION 3 [33 MARKS]

Ques. Solution Explanation Topic
&

Level

3.1.1 Southwest

North 2A Southw est

1A North (4) M&P

L2

3.1.2 Scale refers to the relationship (ratio) between distance on a map
and t he corresponding distance on the ground (reality) 2A Definition (2) M&P
L1

3.1.3 Measure bar = 1,8 cm (Accept 1,7 cm to 1,9 cm)

1,8 cm = 2 km

1,8 cm = 200 000 cm

■ 1 cm = 111 111, 111

■ 1 : 111 000 1A Measure bar

1C km to cm

1S Sim plification

1R Nearest '000 (4) M&P

L3

3.1.4 Speed = ■■■■■■■■

■■■■■

65 km/h = ■■,■ ■■

■■■■■

Time = 18,2 ■■

65 ■■/■

= 0,28....h

= 16,8 min

= 17 minutes

Time of arrival = 14:53 + 17 minutes

= 15:10

1SF Substitution

1M Changing

subject of formule

1A Time in hours

1CA Time in min

1M Add times

1CA Arrival time

(6) M&P

L3

3.1.5 Because of the (Indian) ocean 2R Reason (2) M&P

L4

3.2.1

Steak = 0,454 kg

2A Kilogram (2) M

L1

3.2.2 Salt = $0,5 \times 5$ ml

= 2,5 ml

Black pepper = $0,25 \times 5$ ml

$= 1,25 \text{ ml}$
 Total = $2,5 \text{ ml} + 1,25 \text{ ml}$
 $= 3,75 \text{ ml}$
 1MA Millilitres

1CA Millilitres

11CA Total (3) M
 L2

3.2.3 1 cup broth = 250 ml
 3
 4 cup whipping cream = $0,75 \times 250 \text{ ml}$
 $= 187,5 \text{ ml}$

1
 4 cup sour cream = $0,25 \times 250$
 $= 62,5 \text{ ml}$
 Total = $250 \text{ ml} + 187,5 \text{ ml} + 62,5 \text{ ml}$
 $= 500 \text{ ml} = 0,5 \text{ litres}$
 Statement is valid ☒ O 1MA 250 ml

1MA 187,5 ml

1MA 62,5 ml

1CA Litre
 10 Valid (5) M
 L4 ☒ ☒ A
☒ ☒ A
☒ A
☒ C
☒ S
☒ R
☒ M
☒ A
☒ C
☒ M
☒ CA
☒ ☒ R ☒ M
☒ ☒ A
☒ MA
☒ CA
☒ CA
☒ MA
☒ MA
☒ CA ☒ MA ☒ ☒ A

(EC/NOVEMBER 2020) MATHEMATICAL LITERACY P2 5
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3.2.4 Cooking time
 $= (3 \text{ min} \times 2) + 8 \text{ min} + 1 \text{ min} + 1 \text{ min} + 2 \text{ min} + 20 \text{ min} + 2$
 min
 $= 40 \text{ minutes}$ 1M Multiply by 2
 1MCA Adding all
 times
 1CA Total time (3) M
 L2

3.2.5
 Time for preparation should also be considered.

2A Reason (2) M

L4

QUESTION 4 [20 MARKS]

Ques. Solution Explanation Topic
&

Level

4.1.1 6 seats 2A No of seats

(2) M&P

L1

4.1.2 Raised Seating Area = 112 seats

Flat Seating Area = 85 seats

Difference = 112 – 85

= 27 seats 1A Seats in RSA

1A Seats in FSA

1M Subtract

1CA Difference (4) M&P

L3

4.1.3 P(L-row) = 12

197 1A Numerator

1MCA

Denominator (CA

from 4.1.2) (2) M&P

L2

4.1.4 Walk to A5, then turn right

Walk straight pass the Row D

Third Row (G) third seat

Accept any other relevant responses 1A Straight

1A Turn right

1A 3rd row, 3rd seat

(3) M&P

L4

4.2.1 Volume of drum = $\pi \times \text{radius} \times \text{radius} \times \text{height}$

= $3,142 \times 29 \text{ cm} \times 29 \text{ cm} \times 93 \text{ cm}$

= 245 745, 246 cm³

Litres = 245 745, 246 cm³ ÷ 1 000 cm³

= 245,745 litres

1C mm to cm

1A Radius

1SF Substitution

1CA Volume

1CA Litres (5) M

L3

4.2.2 ■ The volume of advertisement refers to the maximum the
drum can hold. ■■A

■ The calculated volume refers to the entire drum. 2A 1st Reason

2A 2nd Reason (4) M

L4

TOTAL: 100 ■A ■■A

■A

■A

■M

■CA

■A

■A ■A

■MCA ■M ■MCA

■CA

■■A

■A
■SF ■C
■CA
■CA
■■A

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GRADE 11

NOVEMBER 2020

MATHEMATICAL LITERACY P2
(EXEMPLAR)

MARKS: 100

TIME: 2 hours

This question paper consists of 8 pages and an addendum with 4 pages.

imlit2

2 MATHEMATICAL LITERACY P2 (EC/NOVEMBER 2020)

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INSTRUCTIONS AND INFORMATION

Read the following instructions carefully before answering the questions.

1. This question paper consists of FOUR questions. Answer ALL the questions.
2. Use the ADDENDUM with ANNEXURES for the following questions:
ANNEXURE A for QUESTION 2.2,
ANNEXURE B for QUESTION 3.1,
ANNEXURE C for QUESTION 3.2 and
ANNEXURE D for QUESTION 4.1
3. Number the questions correctly according to the numbering system used in this question paper.
4. An approved calculator (non-programmable and non-graphical) may be used, unless stated otherwise.
5. Show ALL calculations clearly.
6. Maps and diagrams are NOT drawn to scale, unless otherwise stated.
7. Indicate units of measurement, where applicable.
8. Round off ALL final answers appropriately according to the given context, unless stated otherwise.
9. Start EACH question on a NEW page.
10. Write neatly and legibly.

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QUESTION 1

1.1 Study the following thermometers and answer the questions that follow.

[Source: [www.http://:VectorStock.com/22441849](http://www.VectorStock.com/22441849)]

1.1.1 Write down the temperature reading in degrees Celsius ($^{\circ}\text{C}$) of thermometer 1.
(2)

1.1.2 Identify the temperature in degrees Fahrenheit ($^{\circ}\text{F}$), if the temperature in degrees Celsius ($^{\circ}\text{C}$) is 35. (2)

1.1.3 Use thermometer 2 and write down the minimum temperature in degrees Fahrenheit ($^{\circ}\text{F}$) as illustrated on the thermometer. (2)

1.1.4 Write the temperature reading in degrees Fahrenheit ($^{\circ}\text{F}$) of thermometer 1 to thermometer 2 as a ratio in its simplest form.
(3)

4 MATHEMATICAL LITERACY P2 (EC/NOVEMBER 2020)

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1.2 The following is a street map of a neighbourhood. Study the street map and answer the questions.

[Source: [www.http://getdrawings.com](http://www.getdrawings.com)]

1.2.1 How many houses are there in this neighbourhood? (2)

1.2.2 Which building is located next to the park? (2)

1.2.3 Which business appears more than all the other businesses? (2)

1.2.4 The bar (graphic or line) scale shows the distance of the bar in real life. Measure the length of the bar in centimetres. (2)

1.2.5 Hence, use your measurement in QUESTION 1.2.4 to explain the scale of this street map. (2)

1.2.6 How many more houses are in the street at the top than the street at the bottom? (2)
[21]

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QUESTION 2

2.1 The following diagram shows a cylindrical lamp shade. Use the information to answer the questions that follow.

Dimensions of the lamp shade:

Height = 31 cm
Diameter = 200 mm

You may use the following:

Circumference of lamp shade = $\pi \times \text{diameter}$

Area of lampshade = $\pi \times \text{radius}^2$ where $\pi = 3,142$

2.1.1 Explain the term circumference in this context. (2)

2.1.2 Determine the radius of the lamp shade in centimetres. (3)

2.1.3 Hence, calculate the circumference of the lamp shade in centimetres. (2)

2.1.4 A worker stated that the height of the lamp shade is 65% more than the diameter of the lamp shade.

Verify, with the necessary calculations, whether the statement is valid or not. (3)

2.1.5 The lamp shade needs to be covered with material.

Calculate the area for the material required if 4,25% wastage must be allowed for seams and overlaps. Give your final answer to the nearest square centimetre. (5)

Height
diameter

6 MATHEMATICAL LITERACY P2 (EC/NOVEMBER 2020)

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2.2 ANNEXURE A shows the instructions of how a floor lamp is assembled. Use ANNEXURE A to answer the questions below.

2.2.1 How many parts are used to assemble the floor shade excluding the tools (Allen keys and wrench)? (2)

2.2.2 Determine the probability of randomly using a set screw to assemble the floor lamp. Write your final answer to 3 decimal places. (3)

2.2.3 Give a reason why the manufacturer included an 'IMPORTANT' instruction to the assembly. (2)

2.2.4 Explain why the manufacturer did not include a bulb in the packaging. (2)

2.2.5 Explain the purpose of the connector. (2)

[26]

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QUESTION 3

3.1 Study the map, ANNEXURE B, that shows an extract of Durban. Answer the following questions that refer to the map, ANNEXURE B.

3.1.1 Give the TWO general directions that will be travelled from the Durban City Centre via Berea to Sydenham. (4)

3.1.2 Explain the term scale. (2)

3.1.3 Use the linear (graphic) scale on the map and rewrite it as a numeric scale as 1 : ... to the nearest thousand. (4)

3.1.4 The distance from The Bluff to Riverside is 18,2 km when using the M4. At what time will you arrive in Riverside if you leave the Bluff at 14:53 at a speed of 65 km/h?

You may use the following:

Speed = 

(6)

3.1.5 The city of Durban did NOT expand to the east. What is the reason for this? (2)

3.2 Refer to the recipe, ANNEXURE C, and answer the questions below. You may use the following:

1 pound (lb) = 0,454 kilogram
1 tablespoon = 15 millilitres
1 teaspoon = 5 millilitres
1 cup = 250 millilitres
1 litre = 1 000 millilitres

3.2.1 How many kilograms of steak is used for the Beef Stroganoff? (2)

3.2.2 Determine the total millilitres of salt and black pepper to be used for the recipe. (3)

3.2.3 A new chef stated that the total amount of ingredients for the broth, whipping cream and sour cream is exactly half a litre. Verify, with the necessary calculations, whether his statement is valid or not. (5)

3.2.4 Determine the maximum cooking time for the Beef Stroganoff. (3)

3.2.5 Explain why the time for cooking is NOT the only time that should be considered for making the Beef Stroganoff. (2)
[33]

8 MATHEMATICAL LITERACY P2 (EC/NOVEMBER 2020)
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QUESTION 4

4.1 ANNEXURE D shows a seating plan of a cinema. Study the seating plan and answer the questions.

4.1.1 Write down the number of seats in Row N. (2)

4.1.2 How many more seats are there in the 'Raised Seating Area' than in the 'Flat Seating Area'? (4)

4.1.3 Determine the probability that someone will be seated in the 'L' row. (2)

4.1.4 You are seated at Seat A17, while your cousin is seated at G7. You need to give your cousin something. Explain how you will walk to reach your cousin if you walk past the stage/screen area. (3)

4.2 The following diagram shows a drum that is advertised as 200 litres. Answer the questions below.

Dimensions of the drum:
Diameter = 580 mm
Height = 93 cm

You may use the following:

Volume of drum = $\pi \times \text{radius} \times \text{radius} \times \text{height}$, where $\pi = 3,142$

1 000 cm³ = 1 litre

4.2.1 Show, with the necessary calculations, that the volume of the drum in litres is different than the advertised 200 litres. (5)

4.2.2 Hence, explain why the two volumes in litres are different. (4)
[20]

TOTAL: 100

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MATHEMATICAL LITERACY P2
ADDENDUM
(EXEMPLAR)

This addendum consists of 5 pages with a 4-page annexure.

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2 MATHEMATICAL LITERACY P2 (ADDENDUM) (EC/NOVEMBER 2020)

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QUESTION 2.2

ASSEMBLE A – FLOOR LAMP

IMPORTANT: Always disconnect the power before installing or replacing a bulb and before cleaning or other maintenance.

Assembly and installation
instructions

1. Attach the base to stem 3, inserting the threaded stem and locator pin through the washer and in the base. Secure the lock washer and hex nut using the wrench (included).
2. Insert the connector on stem 2 into stem 3 and use the large Allen key to tighten the set screws. (Loosen the set screws to adjust the height of the lamp)
3. Insert stem 1 into stem 2 and use the small Allen

key to tighten the set
screw.

4. Attach the diffuser, shade, spacer, and harp to the socket, and secure with socket ring.

5. Install a 100- Watt medium
base bulb (not included).

6. Attach the glass and flat
plate to the harp and
secure with the finial.

7. Plug in the cord and the
dimmer switch is located
on the cord. Floor Lamp

[Source: <https://manualzz.com/doc/19886634/assembly-instructions-for-item>]

(EC/NOVEMBER 2020) MATHEMATICAL LITERACY P2 (ADDENDUM) 3

Copyright reserved Please turn over ANNEXURE B

QUESTION 3.1

4 MATHEMATICAL LITERACY P2 (ADDENDUM) (EC/NOVEMBER 2020)

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QUESTION 3.2

RECIPE FOR BEEF STROGANOFF

For the Beef Stroganoff:

x 1 lb (pound) top sirloin steak thinly sliced into strips

x 2 (tbsps.) tablespoon olive oil

x 2 (tbsps.) tablespoon butter

x ■

■ medium onion, finely chopped

x ■

■ lb (pound) brown mushrooms, thickly sliced

x 1 garlic clove minced

x 1 (tbsp.) tablespoon all-purpose flour

x 1 cup beef broth

x ■

■ cup heavy whipping cream

x ■

■ cup sour cream

x 1 (tbsp.) tablespoon Worcestershire sauce

x ■

■ (tsp.) teaspoon Dijon mustard

x ■

■ (tsp.) teaspoon salt

x ■

■ (tsp.) teaspoon black pepper

Instructions

1. Place a large deep pan over medium-high heat. Add 2 tbsps. oil and once the oil is very hot, add thinly sliced beef strips in a single layer, cooking 3 minutes per side without stirring. Cook until just browned and no longer red. Remove beef to a plate and cover to keep warm.

2. Add 2 tbsps. butter, chopped onion and sliced mushrooms. Sauté 6 to 8 minutes or until liquid has evaporated and onions and mushrooms are soft and lightly browned.

3. Add 1 minced garlic clove and sauté 1 minute until fragrant. Add 1 tbsp. flour and sauté another minute stirring constantly.

4. Pour in 1 cup of beef broth, scraping any bits from the bottom of the pan. Then add

■ cup of whipping cream and simmer another 1 to 2 minutes or until slightly thickened.

5. Stir a few tablespoons of the sauce into ■ cup of sour cream to temper it so the sour cream does not curdle. Then add it to the pan while stirring constantly.

6. Stir in 1 tbsp. Worcestershire, ■ tsp Dijon mustard, and season with ■■ tsp salt and ■■ tsp pepper, or season to taste and continue simmering for 20 minutes until sauce is creamy. Add beef with any accumulated juices back to the pan and bring just to a simmer for 2 minutes until beef is heated through.

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QUESTION 4.1

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MATHEMATICAL LITERACY P2
MARKING GUIDELINE
(EXEMPLAR)

MARKS: 100

This marking guideline consists of 5 pages .

Symbol Explanation

M Method

M/A Method with Accuracy

MCA Method with Consistent Accuracy

CA Consistent Accuracy

A Accuracy

C Conversion

S Simplification

RT/RG/RM Reading from a table OR Reading from a graph OR Read from map

F Choosing the correct formula

SF Substitution in a formula

J Justification

P Penalty, e.g. for no units, incorrect rounding off etc.

R Rounding Off OR Reason
AO Answer only
NPR No penalty for rounding
2 MATHEMATICAL LITERACY P2 (EC/NOVEMBER 2020)
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QUESTION 1 [21 MARKS]

Ques. Solution Explanation Topic

&

Level

1.1.1 Temperature reading = 30 °C 2A Correct reading (2)

M

L1

1.1.2 °F = 100 °F 2A °F (2) M

L1

1.1.3 Minimum °Fahrenheit = minus 42°F OR - 42°F

2A Min °F (2) M

L1

1.1.4 Thermometer 1 : Thermometer 2

=

90 : 20

= 9 : 2 1RD Both values correct

1M Correct order 1CA Simplified ratio (3) M

L1

1.2.1 Number of houses = 18 2A Correct number of houses (2) M

L1

1.2.2 Hospital 2A Correct building (2) M

L1

1.2.3 Café 2A Correct business (2) M

L1

1.2.4 Length of the bar = 1,5 cm 2A Correct length (2)

M

L1

1.2.5 1,5 cm on the map represents 50 yards in reality CA from 1.2.4

2A Explanation (2) M

L1

1.2.6 3 houses 2A Number of houses more (2) M

L1

[21] 99A

99A

99A

99A

99A

99A

99A

99A 9RD

9M

9CA

99A

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QUESTION 2 [26 MARKS]

Ques. Solution Explanation Topic

&

Level

2.1.1 Circumference is the distance or length around the lamp shade 99A 2A Explanation (2) M
L1

2.1.2 Radius = ■■■■

■■■

= ■■■

■

= 10 cm 1C mm to cm

1M Divide by 2 1CA Radius (3) M

L2

2.1.3 Circumference of lamp shade = ■ × d i a m e t e r
= 3,142 × 20 cm
= 62,84 cm CA from 2.1.1

1SF Substitution

1MCA

Circumference (2) M

L2

2.1.4 Height of lamp shade = 20 cm × 1,65
= 33 cm

Statement invalid 1M Multiply by

1,65

1A Height

1O Invalid (3) M

L4

2.1.5 Area of lampshade = ■ × r a d i u s × r a d i u s
= 3,142 × 10 cm × 10 cm
= 314, 2 cm²

Area including wastage = 314, 2 cm² × 1,0425
= 327,5535 cm²
§ 328 cm² CA from 2.1.1

1SF Substitution

1MCA Area of lamp shade

1M Multiply by

1,0425

1CA Area including

wastage

1R Nearest cm² (5) M

L3

2.2.1 Number of parts = 22 parts 2A Number of parts
(2) M&P

L2

2.2.2 P(set screw) = ■

■■■

= 0,090909....

§ 0,091 CA from 2.2.1

1A Number of set

screws

1MCA

1R 3 dec places (3) P

L2

2.2.3 To prevent shocking

OR

For safety purposes

Accept any other relevant reasons 2R Reason (2) M&P

L4

2.2.4 Bulbs can break/damage 2R Explanation (2) M&P
L4

2.2.5 To connect Stem 2 with Stem 3 2A Explanation (2) M&P
L4 9C

9M

9CA

9SF

9MCA

9M

9A

9O

9SF

9MCA

9M

9CA

9R

99A

9A

9MCA

9R

99R

99R

99A

4 MATHEMATICAL LITERACY P2 (EC/NOVEMBER 2020)

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QUESTION 3 [33 MARKS]

Ques. Solution Explanation Topic

&

Level

3.1.1 Southwest

North 2A Southwest

1A North (4) M&P

L2

3.1.2 Scale refers to the relationship (ratio) between distance on a map
and the corresponding distance on the ground (reality) 2A Definition (2) M&P

L1

3.1.3 Measure bar = 1,8 cm (Accept 1,7 cm to 1,9 cm)

1,8 cm = 2 km

1,8 cm = 200 000 cm

■ 1 cm = 111 111, 111

■ 1 : 111 000 1A Measure bar

1C km to cm

1S Simplification 1R Nearest '000 (4) M&P

L3

3.1.4 Speed = ■■■■■■■■

■■■■■

65 km/h = ■■, ■■■

■■■■■

Time = ■■, ■■■

■■ ■■/■

= 0,28 ...h

= 16,8 min

= 17 minutes

Time of arrival = 14:53 + 17 minutes

= 15:10

1SF Substitution 1M Changing subject of formule 1A Time in hours 1CA Time in min 1M Add
times

1CA Arrival time
(6) M&P

L3

3.1.5 Because of the (Indian) ocean 2R Reason (2) M&P
L4

3.2.1
Steak = 0,454 kg
2A Kilogram (2) M
L1

3.2.2 Salt = $0,5 \times 5$ ml
= 2,5 ml
Black pepper = $0,25 \times 5$ ml
= 1,25 ml
Total = 2,5 ml + 1,25 ml
= 3,75 ml

1MA Millilitres 1CA Millilitres 11CA Total (3) M
L2

3.2.3 1 cup broth = 250 ml
■
■ cup whipping cream = $0,75 \times 250$ ml
= 187,5 ml
■
■ cup sour cream = $0,25 \times 250$
= 62,5 ml
Total = 250 ml + 187,5 ml + 62,5 ml
= 500 ml = 0,5 litres
Statement is valid 9O 1MA 250 ml
1MA 187,5 ml

1MA 62,5
ml

1CA Litre
1O Valid (5) M
L4 99A

99A

9A

9C

9S

9R

9M

9A

9C

9M

9CA

99R 9M

99A

9MA

9CA

9CA

9MA

9MA

9CA 9MA 99A

(EC/NOVEMBER 2020) MATHEMATICAL LITERACY P2 5

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3.2.4 Cooking time = (3 min
 × 2) + 8 min + 1 min + 1 min + 2 min + 20 min + 2
 min
 = 40 minutes 1M Multiply by 2 1MCA Adding all
 times

1CA Total time (3) M
 L2

3.2.5 Time for preparation should also be considered. 2A Reason (2) M
 L4

QUESTION 4 [20 MARKS]
 Ques. Solution Explanation Topic
 &
 Level

4.1.1 6 seats 2A No of seats (2) M&P
 L1

4.1.2 Raised Seating Area = 112 seats Flat Seating Area = 85 seats

Difference = 112 – 85
 = 27 seats 1A Seats in RSA
 1A Seats in FSA 1M Subtract

1CA Difference (4) M&P
 L3

4.1.3 P(L-row) = ■■■
 ■■■■ 1A Numerator
 1MCA Denominator (CA
 from 4.1.2) (2) M&P
 L2

4.1.4 Walk to A5, then turn right
 Walk straight pass the Row D
 Third Row (G) third seat
 Accept any other relevant responses 1A Straight
 1A Turn right
 1A 3rd row, 3rd seat
 (3) M&P

L4
 4.2.1 Volume of drum = ■ × radius × radius × height

$$= 3,142 \times 29 \text{ cm} \times 29 \text{ cm} \times 93 \text{ cm}$$

$$= 245\,745,246 \text{ cm}^3$$

Litres = 245 745, 246 cm³ ÷ 1 000 cm³
 = 245,745 litres

1C mm to cm
 1A Radius
 1SF Substitution
 1CA Volume
 1CA Litres (5) M
 L3

4.2.2 x The volume of advertisement refers to the maximum the
 drum can hold. 99A
 x The calculated volume refers to the entire drum. 2A 1st Reason
 2A 2
 nd Reason (4) M
 L4

TOTAL: 100 9A 99A
 9A

9A
9M
9CA
9A
9A 9A
9MCA 9M 9MCA
9CA
99A
9A
9SF 9C
9CA
9CA
99A